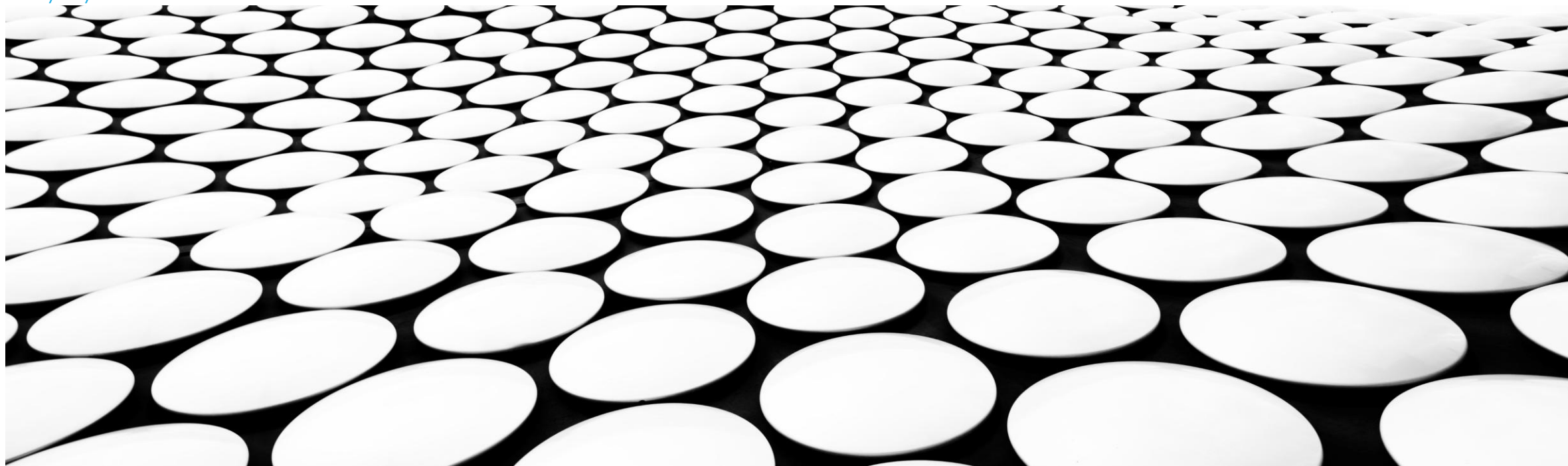


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# CHAPTER 6 OF BECOMING A DATA HEAD; EXPLORATORY DATA ANALYSIS

BY SOROUSH HOSSEINI

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## CHAPTER 6: EXAMINE THE PROBABILITIES

### The Language of Uncertainty

*"Many people's notion of probability is so impoverished that it admits only two values: 50-50 and 99%, tossup or essentially certain."*

— John Allen Paulos



## What We'll Cover:

- Understanding probability as the language of uncertainty
- The rules and notation of probability
- Recognizing common traps and fallacies
- Bayes' Theorem and its applications
- Making probabilities meaningful in the workplace

Probability is counter-intuitive; your gut feelings often mislead you.

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# WHY PROBABILITY MATTERS

1. There is variation in all things
2. Variation creates uncertainty
3. Probability helps us manage uncertainty
4. Without probability, we're guessing blindly

*"Probability and statistics are tools to help us manage uncertainty."*

# PROBABILITY IS HARD!

- Experts disagree on its interpretation
- Intuition often fails us
- Every problem is a brainteaser
- Most people don't have a grasp of probability

*"Probability will continue to confound and confuse. Perhaps the best we can do to combat this is point out some common traps."*

- 
- 1% of laptops infected with a virus
  - Test is 99% accurate (positive if infected, negative if not)
  - Your laptop tests POSITIVE

**Question: What is the probability your laptop actually has the virus?**



## The Surprising Answer:

**Common Guess:** 99%

**Correct Answer:** 50%

*"Surprised? Most people are. The answer isn't intuitive."*

This is the "Confusion of the Inverse"

## WHY WE GET IT WRONG?

- We asked for:  $P(V \mid +)$  = Probability of virus given positive test
- We knew:  $P(+ \mid V)$  = Probability of positive test given virus
- These are **NOT** the same!

*"The probabilities  $P(V \mid +)$  and  $P(+ \mid V)$  are not the same, but they are linked through Bayes' theorem."*

- 
- A positive test is not as definitive as it seems
  - Base rates matter (1% infection rate)
  - You get as many false positives as true positives
  - *"A positive test... is not as definitive as it seems."*

**Conclusion:** Don't jump to conclusions based on a single test!

# WHAT IS PROBABILITY?

Probability measures the likelihood an event will occur; Measured between 0 and 1 (or 0% to 100%)

0 = Impossible, 1 = Certain

## Examples:

$P(\text{Roll a 7 on a die}) = 0$

$P(\text{Roll less than 7}) = 1$

$P(\text{Flip heads}) = 1/2 = 50\%$

$P(\text{Draw a spade}) = 13/52 = 25\%$

# CONDITIONAL PROBABILITY

- Probability of event A **given** event B has occurred
- Notation:  $P(A \mid B)$  — read as "probability of A given B"

## Examples:

- $P(\text{Alex late}) = 5\%$
- $P(\text{Alex late} \mid \text{flat tire}) = 100\%$
- $P(\text{Alex late} \mid \text{traffic jam}) = 50\%$

*"The probability of an event occurring depends heavily on the event, or events, preceding it."*

# INDEPENDENT EVENTS

- When one event doesn't affect another, they're independent
- $P(A | B) = P(A)$  when independent
- $P(B | A) = P(B)$  when independent

**Example:** Coin flip and drawing a card

- $P(\text{Spade} | \text{Heads}) = P(\text{Spade})$
- The deck doesn't care what happened with the coin

*"There is no dependency between the two events."*

# TWO THINGS HAPPENING TOGETHER – THE MULTIPLICATIVE RULE

- $P(A, B) = P(B) \times P(A \mid B)$
- For independent events:  $P(A, B) = P(A) \times P(B)$

**Example:** Alex and Jordan late

- $P(\text{Jordan late}) = 10\%$
- $P(\text{Alex late} \mid \text{Jordan late}) = 20\%$
- $P(\text{Both late}) = 10\% \times 20\% = 2\%$
- *"The chance of any two events happening together cannot be greater than either event happening by itself."*



## The Overlap Principle:

The overlap (intersection) can never be larger than the smaller circle

More conditions = lower probability

$$P(A, B) \leq P(A) \text{ and } P(A, B) \leq P(B)$$

**Example:** Can't be both late more often than Alex is late

*"The chance of any two events happening together cannot be greater than either event happening by itself."*

# ONE THING OR THE OTHER – THE ADDITIVE RULE

- $P(A \text{ or } B) = P(A) + P(B) - P(A, B)$
- Subtract the overlap to avoid double-counting

**Example:** Alex or Jordan late

- $P(\text{Alex late}) = 5\%$
- $P(\text{Jordan late}) = 10\%$
- $P(\text{Both late}) = 2\%$
- $P(\text{Alex OR Jordan late}) = 5\% + 10\% - 2\% = 13\%$

*"We can't just add both probabilities, because that would double count the days when we were both late."*

# WHY WE MUST SUBTRACT OVERLAP – THE DICE EXAMPLE

- Without subtraction, probability can exceed 1
- $P(\text{Roll} > 2) = 4/6$
- $P(\text{Roll is odd}) = 3/6$
- $4/6 + 3/6 = 7/6 > 1$  (impossible!)
- Overlap (numbers 3 and 5) =  $2/6$
- Correct answer =  $4/6 + 3/6 - 2/6 = 5/6$

*"The challenging part for some when dealing with the probability of multiple events is subtracting the overlap."*

## The Linda Problem – Sam

- Sam is 29, reserved, very bright
- Majored in economics
- Volunteered at statistical consulting center
- Taught himself Python programming

**The Question:** Which is more probable?

Sam lives in Ohio

Sam lives in Ohio and works as a data scientist

*"This is a rebranding of the popular Linda problem."*

## The Linda Problem – The Answer

**Correct Answer: #1** (Sam lives in Ohio)

$P(\text{Ohio and Data Scientist}) \leq P(\text{Ohio})$

Adding conditions always reduces probability

More details = narrower possibilities

*"The more 'and' qualifications you add to any statement, the more you narrow the possibilities."*



## The Linda Problem – The Common Mistake:

People read #2 as: "What is  $P(\text{Ohio} \mid \text{Data Scientist})$ ?"

This is **NOT** the same as  $P(\text{Ohio AND Data Scientist})$

$P(\text{Ohio} \mid \text{Data Scientist})$  can be high

$P(\text{Ohio AND Data Scientist})$  is low

*"The language difference between 'and' and 'given' matters."*

## TRAP 1 – ASSUMING INDEPENDENCE

- Banks assumed mortgage defaults were independent
- Model:  $P(A \text{ and } B) = P(A) \times P(B)$
- This **underestimated** risk significantly

### Reality:

- Defaults were dependent events
- One default → property values drop → more defaults
- Like dominoes, not isolated events

*"Assuming independence when events are not independent is a mistake made again and again."*

# THE BOARDROOM MISTAKE

Three risky projects with failure probabilities:

A: 50%, B: 25%, C: 10%

Wrong math (assuming independence):

$50\% \times 25\% \times 10\% = 1.25\%$  all fail

**This is wrong!**

**Reality:**

- Projects share common risks
- Economic downturn, pandemic affect all
- True probability of all failing is **MUCH HIGHER**

*"By wrongfully assuming independence, they are underestimating the probability all projects will fail next year."*

## TRAP 2 – THE GAMBLER'S FALLACY

- False belief that past outcomes affect future probabilities
- "This slot machine is due to hit"
- "Heads came up 10 times, so tails is more likely now"

### Reality:

- Events are independent
- $P(\text{Heads}) = 50\%$  **EVERY TIME**
- 10 heads in a row? Next flip still 50%
- *"On the flip side, some events are independent but not viewed that way. This creates a different kind of risk that casinos thrive on."*

## TRAP 3 – CONFUSION OF THE INVERSE

$P(A \mid B) \neq P(B \mid A)$  (they are not equal)

### Example:

- $A$  = "Lives in New York State"
- $B$  = "Lives in New York City"
- $P(A \mid B) = 1$  (every NYC resident is in NY State)
- $P(B \mid A) \approx 0.4$  (only 40% of NY State residents live in NYC)

*"Swapping the dependencies and assuming  $P(A \mid B) = P(B \mid A)$  is an error so common it was given a name... Confusion of the Inverse."*

# WHAT IS BAYES' THEOREM?

- $P(A \mid B) \times P(B) = P(B \mid A) \times P(A)$
- Relates two inverse conditional probabilities
- Helps us flip the condition

*"Bayes' theorem, which dates back to the 1700s, is a clever way to work with conditional probabilities."*

# REAL-WORLD APPLICATIONS OF BAYES' THEOREM

## Medical screening:

Know:  $P(+ \text{ test} \mid \text{Cancer})$

Want:  $P(\text{Cancer} \mid + \text{ test})$

## Legal evidence:

Know:  $P(\text{Evidence} \mid \text{Guilty})$

Want:  $P(\text{Guilty} \mid \text{Evidence})$

## Spam filtering:

Know:  $P(\text{Words} \mid \text{Spam})$

Want:  $P(\text{Spam} \mid \text{Words})$

*"Bayes' theorem enables us to relate the conditional probability of two events."*

# THE VIRUS TEST SOLVED

$$P(V \mid +) \times P(+) = P(+ \mid V) \times P(V)$$

Known:

- $P(V) = 1\%$  (infection rate)
- $P(+ \mid V) = 99\%$  (test sensitivity)
- $P(+) = 1.98\%$  (overall positive rate)

Calculate:

- $P(V \mid +) = (99\% \times 1\%) / 1.98\%$
- $P(V \mid +) = 50\%$

*"The probability your laptop has the virus given the positive test is 50%."*

# CALIBRATION – PROBABILITIES SHOULD HAVE MEANING

- Calibration = events occur as often as you say they will
- "I have 70% confidence" → right 70% of the time
- "90% chance of success" → succeeds 90% of the time

## Poor Calibration Example:

- Lawyer says 90% win rate
- Actually wins only 60% of cases

## Overconfident and poorly calibrated

*"Calibration measures whether, over the long run, events occur about as often as you say they're going to occur."*

# RARE EVENTS AND MULTIPLYING PROBABILITIES

Rare events happen:

"1 in a million" = 7,800 people/day (7.8B population)

Don't dismiss them as impossible

**Don't multiply unnecessarily:**

You can make anything seem improbable

"The probability you are reading this exact line... It's clear we were destined for each other!"

*"Rare events can, and do, happen."*

## Three Golden Rules:

- **Be careful assuming independence**
  - Many events are connected
  - False independence → underestimating risk
- **Know that all probabilities are conditional**
  - Every probability has hidden assumptions
  - Don't swap dependencies:  $P(A|B) \neq P(B|A)$
- **Ensure the probabilities have meaning**
  - Check calibration
  - Recognize rare events do happen

*"Probability is hard. But a big part in learning a new topic is respecting that things can go wrong."*